

Fatty acid profile of human saliva: a possible indicator of dietary fat intake.

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OBJECTIVE: Since conventional food questionnaires are not precise in assessing the dietary fatty acids, the purpose of this study was to determine the relationship between the salivary fatty acid profile and the alimentary habits of two different groups in an attempt to develop a more reliable way to determine the lipidic intake. DESIGN: Twenty adults of both sexes, with mixed (M) or vegetarian (V) diets were studied. Data about the fat intake were obtained by means of a Food Frequency Questionnaire (FFQ) and the presence of the main salivary fatty acids was determined by gas chromatography. RESULTS: A greater salivary concentration of alpha-linolenic acid (18:3 n-3) (2.82) was found in V than in M subjects (1.65) ($p = 0.001$), whilst arachidonic acid (20:4 n-6) was lower in V (3.93) than in M (4.52) ($p = 0.045$). The same difference regarding arachidonic acid was observed in the dietary fatty acid intake, also showing a significant correlation between its dietary and salivary levels in vegetarian subjects. CONCLUSIONS: These results show that salivary arachidonic acid, relevant for their eicosanoid production related to the tumourigenesis process and cardiovascular diseases, is influenced by dietary fats.